

Name: C. Venkatesh

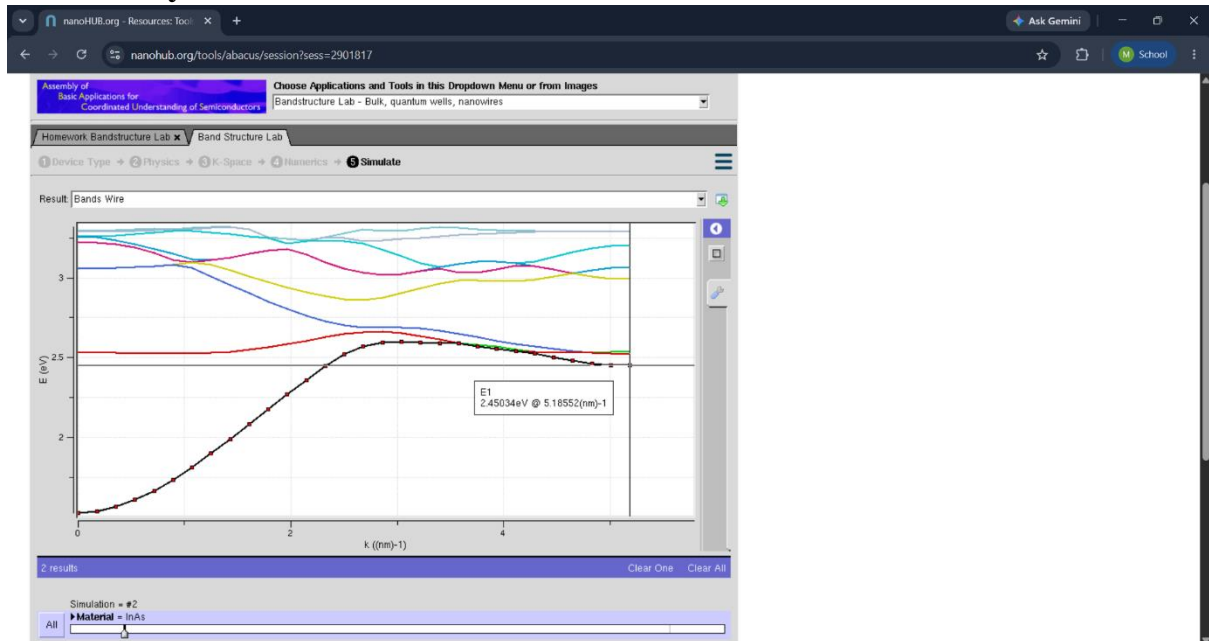
Course: Nanoelectronics

Reg no: 192312325

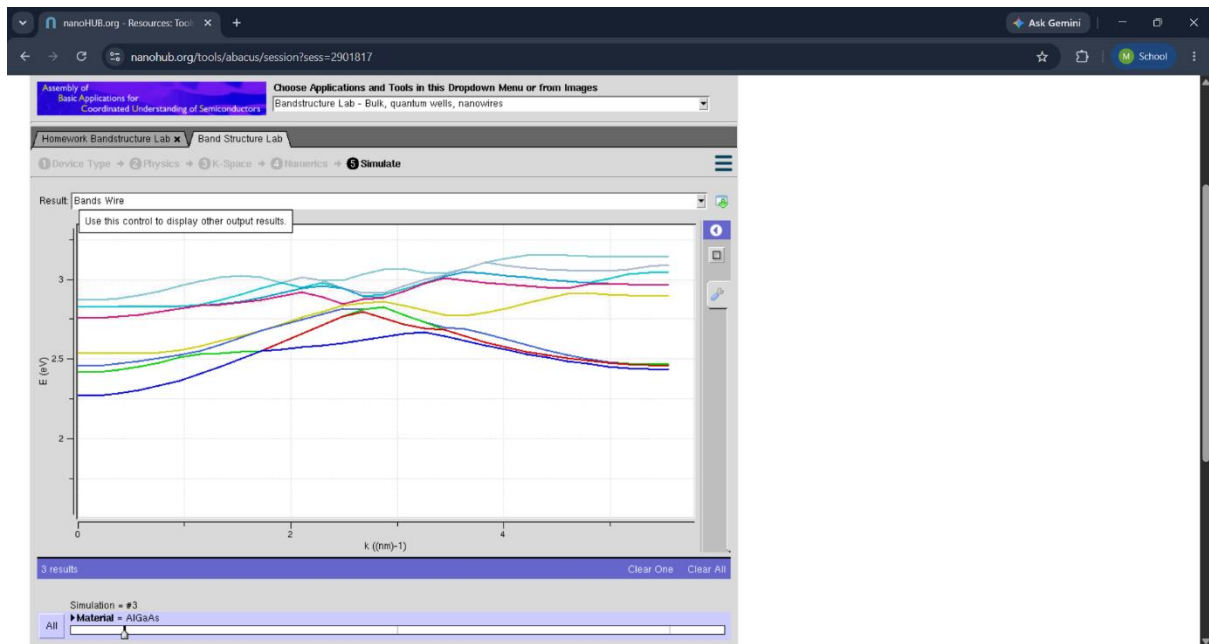
Course code: ECA3102

Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

Case Study 1: Simulate the band-structure of InAs circular nanowire

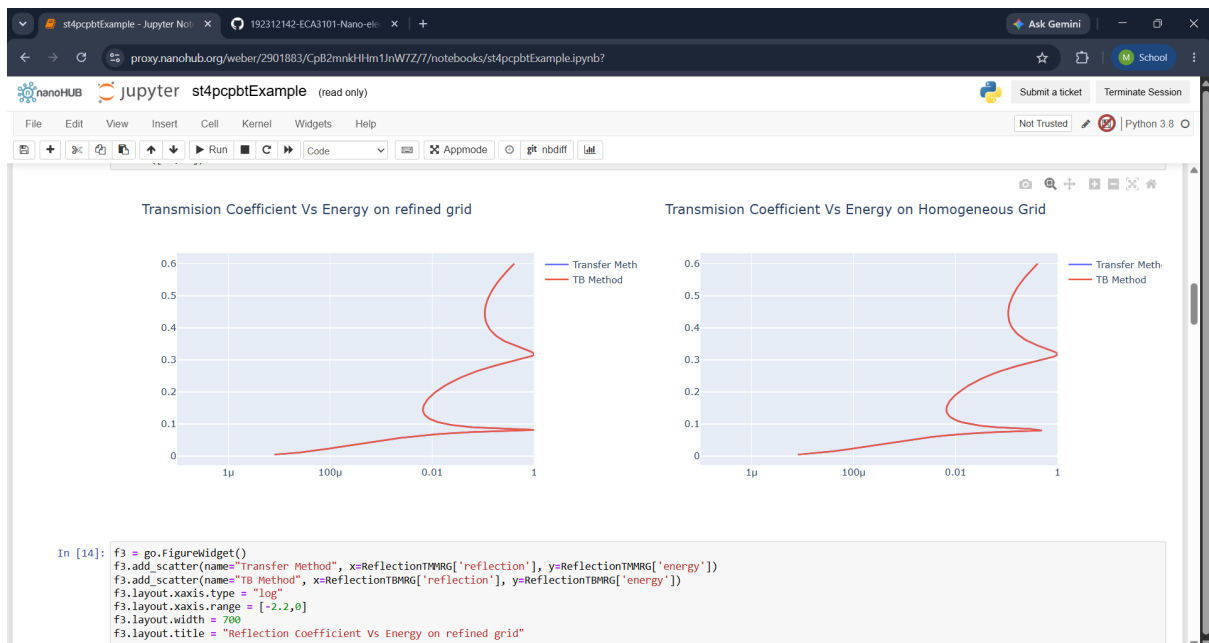


Case Study 2: Simulate the band-structure of AlGaAs circular nanowire

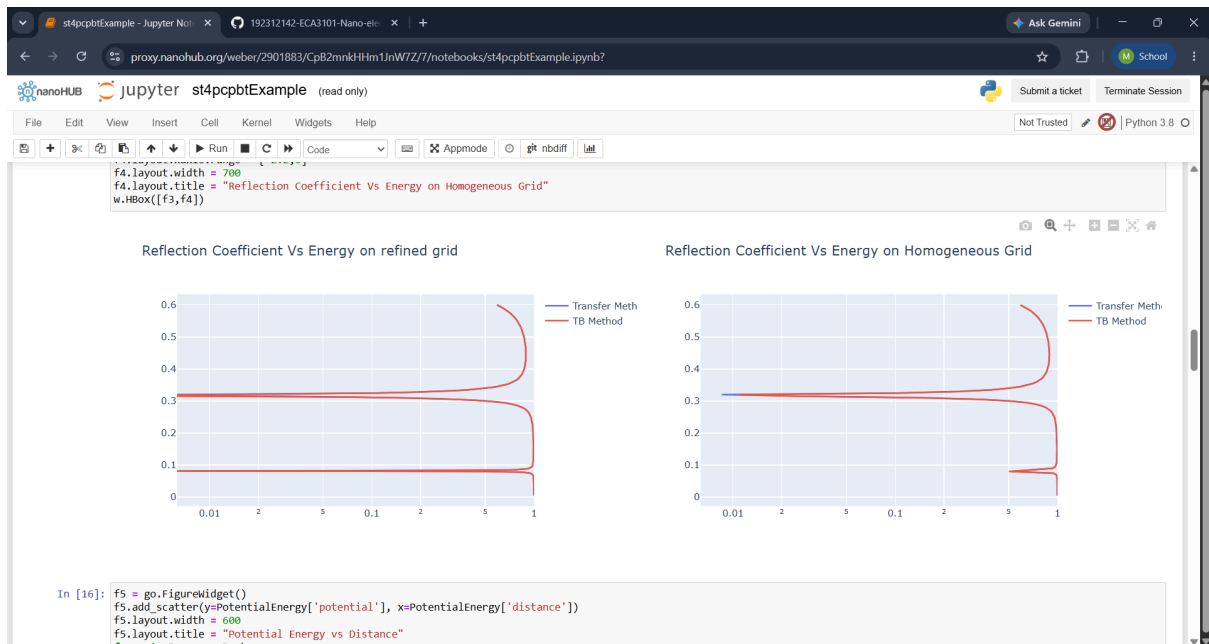


Exp-6 Demonstration of quantum tunnelling

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid



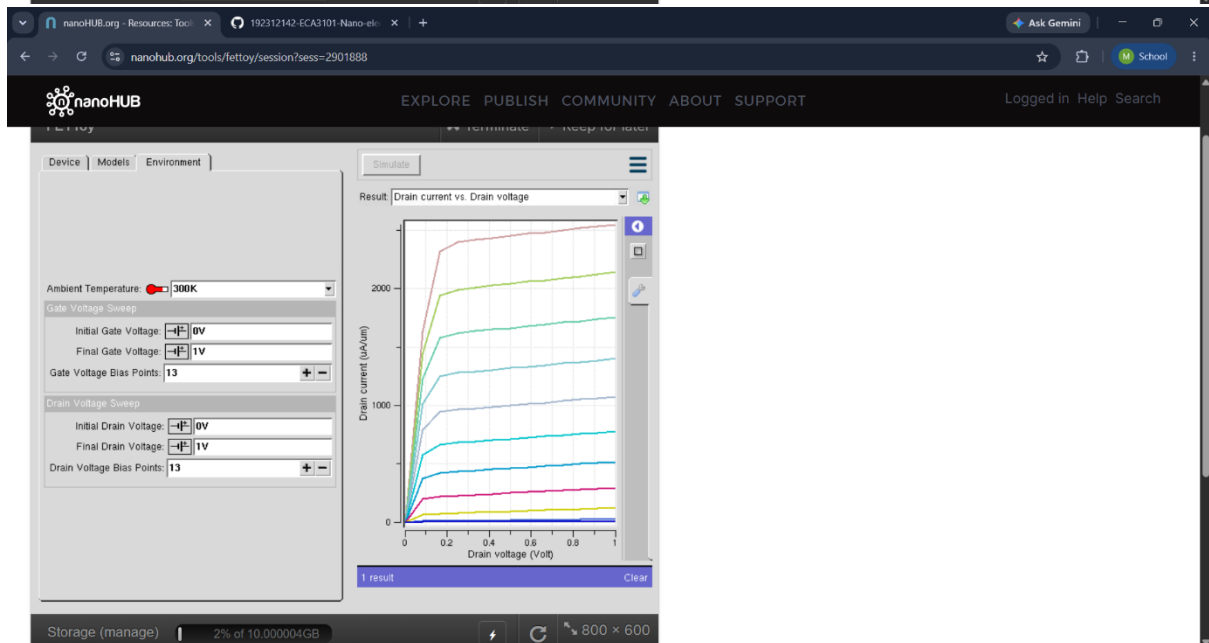
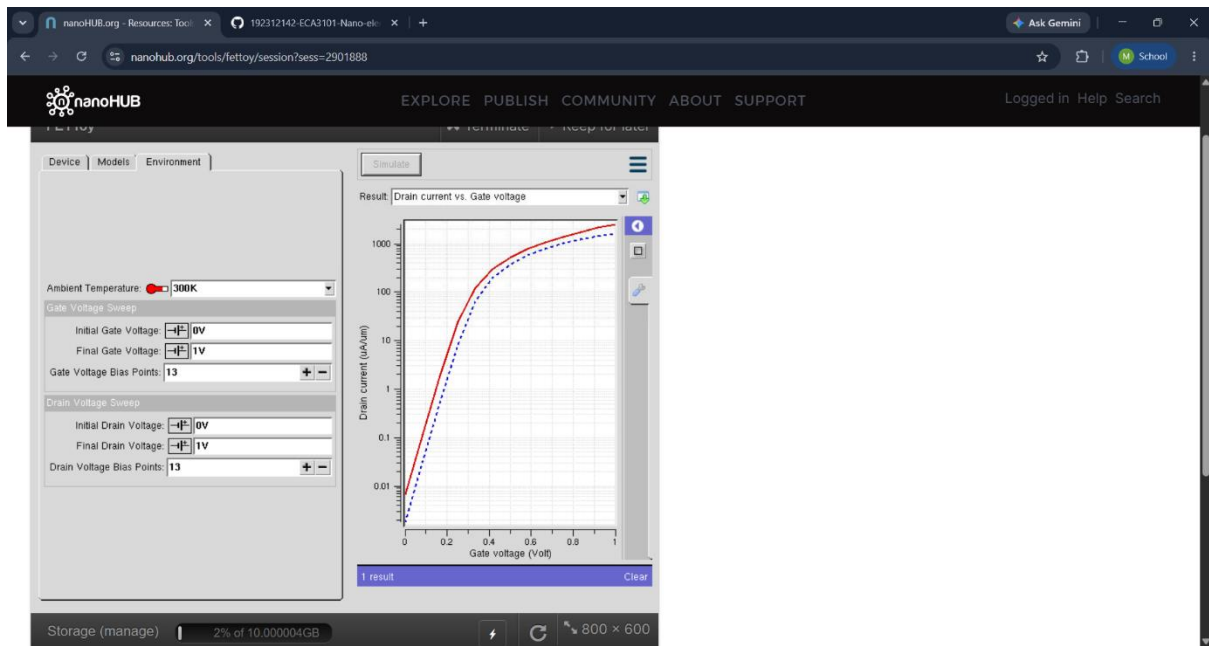
Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid



Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub

Case 1: Increase in V_G :

- **Observation:** As V_G increases, the drain current (I_D) also increases exponentially due to the MOSFET entering the saturation region earlier.



Case 2: Variation in VD :

- **Observation: Drain current (I_D) saturates for higher VD at a fixed VG**

